1) g) $S \to aA\{xx\}$ $\quad$ $S \to aAF \to acBx \to acbSc \to aAFc \to aAc$

$A \to bA \mid cBx \mid \varepsilon \Rightarrow$

$B \to bSc$

$S \to aAF$

$F \to xxF \mid \varepsilon$

$A \to bA \mid cBx \mid \varepsilon$

$B \to bSc$

1) F: first{ε} = ∅, first{xxF} = {x} } ∅

A: first{bA} = {b}, first{cBx} = {c}, first{ε} = ∅ } ∅

2) $F \to \varepsilon$, $F \to xxF$

A: бен

3) F: ε => ε

first{F} = {X}, follow{F} = ∅ } ∅

A: ε => ε :

first{A} = {b,c}, follow{A} = {x,c} } ≠∅ => непрям

i) $S \to bS \mid aAB$ $\quad$ $S \to aAB \to aAcbB \to$

$A \to bcA \mid ccA \mid \varepsilon$

$B \to cbB \mid \varepsilon$

1) S: first{bS} = {b}, first{aAB} = {a}

A: {b} ∩ {c} = ∅

B: {c} ∩ {∅} = ∅

2) бен

3) first{S} = {a,b}, follow{S} = ∅ } ∅

first{A} = {b,c}, follow{A} = {c} } c ≠ 0 => непр.

j) $S \to aASb \mid cfAd$ $\quad$ $S \to aASb \to aAcfAd$

$A \to bA \mid c \mid \varepsilon$

1) S: {a} ∩ {c} = ∅

A: {b} ∩ c ∩ ∅ = ∅

2) бен

3) first{S} = {a,c}, follow{S} = {b} } ∅

first{A} = {b,c}, follow{A} = {c} } c ≠ 0 => непр

опечатка

2) a) $S \to Ad$ $\quad$ $Ad \to BNd \to BaBNd \to baBd$

$A \to B\{aB\} \Rightarrow$ $\quad$ $Ad \to BNd \to$

$B \to b \mid \varepsilon$

$S \to Ad$

$A \to BN$

$N \to aBN \mid \varepsilon$

$B \to b \mid \varepsilon$

B: first{b} = b, first{ε} = ∅ } ∅

2) бен

3) first{B} = {b}, Follow{B} = {a,d} } ∅

N: first{aBN} = a, first{ε} = ∅ } ∅

2) бен

N: first{N} → {a}, follow{N} → {∅} } ∅

} прям

b) $S \to aA \mid bB$ $\quad$ $S \to aA \to acS \to$

$A \to cS \mid \varepsilon \Rightarrow$

$B \to \{, b\}$

$S \to aA \mid bB$

$A \to cS \mid \varepsilon$

$B \to , bB \mid \varepsilon$

1) $first\{aA\} \cap first\{bB\} = a \cap b = \varnothing$ 2) был   3) $\left.\begin{matrix} first\{A\} = c \\ follow\{A\} = \varnothing \end{matrix}\right\} \varnothing$

$c \cap \varnothing = \varnothing$

$, \cap \varnothing = \varnothing$

$\left.\begin{matrix} first\{B\} = , \varnothing \\ follow\{B\} = \varnothing \end{matrix}\right\} \varnothing$ прин

c) $S \to aSb \mid A$
$A \to b \;\{b\}$ $\Rightarrow$ $S \to aSb \mid A$
$A \to bA \mid \varepsilon$

1) $\left.\begin{matrix} first\{aSb\} = a \\ first\{A\} = b \end{matrix}\right\} \varnothing$ 2) был   3) $\left.\begin{matrix} first\{S\} = \{a, b\} \\ follow\{S\} = \{b\} \end{matrix}\right\} = b$ $\Rightarrow$ непр

$\left.\begin{matrix} first\{bA\} = b \\ first\{\varepsilon\} = \varnothing \end{matrix}\right\} \varnothing$

опечатка?

d) $S \to Ad$
$A \to B\{aB\}B \mid \varepsilon$ $\Rightarrow$ $S \to Ad$
$B \to b \mid \varepsilon$

$A \to BNB \mid \varepsilon$
$N \to aBN \mid \varepsilon$
$B \to b \mid \varepsilon$

1) $\left.\begin{matrix} first\{BNB\} = b \\ first\{\varepsilon\} = \varnothing \end{matrix}\right\} \varnothing$ 2) был   3) $\left.\begin{matrix} first\, A = b \\ follow\, A = \varnothing \end{matrix}\right\} \varnothing$

$\left.\begin{matrix} first\{aBN\} = a \\ first\{\varepsilon\} = \varnothing \end{matrix}\right\} \varnothing$ $\left.\begin{matrix} first\, N = a \\ follow\, N = a \end{matrix}\right\} a \Rightarrow$ непр

$\left.\begin{matrix} first\{b\} = b \\ first\{\varepsilon\} = \varepsilon \end{matrix}\right\} \varnothing$

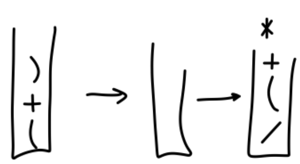
e) $S \to aAS \mid bAS \mid \varepsilon \mid cB$ $S \to aAS \to aAaA \to$
$A \to a \mid \varepsilon \mid bB$
$B \to cBB \mid \varepsilon$

1) был   2) был   3) $\left.\begin{matrix} first\, S = \{a, c, b\} \\ follow\, S = \varnothing \end{matrix}\right\} \varnothing$

$\left.\begin{matrix} first\, A = \{a, b\} \\ follow\, A = a \end{matrix}\right\} \neq \varnothing \Rightarrow$ непр

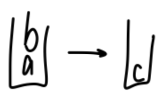
1. d) $(a+b)/(c+a*b)$

$ab+cab*+/$



2. a) $ab*c+$

$a*b+c$



f) abca and or and

| a, c, b, a | and c => | a and c, b, a | b or (c and a) => | b or (c and a), a | => a and (b or (c and a))

g) 2X+2X*<

2+X<2*X

| X, 2 | → | X, 2 | → | < |

3) a) x y * x y / +

| y, x | | y, x, x·y | | x/y, x·y |

$(x/y)+(x\cdot y)=8/2+8\cdot 2=4+16=20$

b) a 2 + b / b 4 * +

| 2, a | | b, (a+2) | | 4, b, (a+2)/b | | b·4, (a+2)/b |

$b\cdot 4+(a+2)/b=3\cdot 4+6/3=\boxed{14}$

c) ab not and a or not

| b, a | → | not b, a | | a, a and (not b) |

not(a or (a and (not b)) = not(1 || (1 & 0)) = not 1 = 0

$\boxed{\text{False}}$

d) x y * 0 > y 2 x − < and

| y, x | | 0, x·y | | x, 2, y, x·y>0 | | 2−x, y, x·y>0 | | 2−x<y, x·y>0 |

(2−x<y) and (x·y>0)

(2−1<1) and (1·1>0)

0 and 1

$\boxed{\text{False}}$

[ ] - применение полиза к выражению

4) i) do {x = y; y = 2*y;} while (x<k);

[x = y;] → x y = ;

[y = 2*y;] → y 2 y * = ;

x = y; y = 2*y; → x y = ; y 2 y * = ;

[x<k] → x k <

L0: S ;
if M goto L0 ;
L0 : S ; if M goto L0 ;
E' p1 !F

безусл перех

M1 : x y = ; y 2 y * = ; x k < ~~M2~~ 14 !F ~~M1~~ ! ~~x~~ 15 M2 :

j) S = 0; for (i = 1; i <= k; i = i+1) S += i · i;

S = 0 ; i = 1 ; M1 : if (!(i <= k)) goto M2 ; S += i · i ; i = i+1; goto M1; M2:

$\underbrace{i = 1}_{E_1 = i\,1 = 1;}$ $\underbrace{i <= k}_{E_2 = i\,k <=}$ $\underbrace{S += i \cdot i}_{S\,S\,i\,i\,*\,+\,=;}$ $\underbrace{i = i+1}_{i\,i\,1\,+\,=;}$

S0 = ; i1 = ; M1 : ik <= ~~M2~~ 24 !F SS i i * + = ; i i 1 + = ; M1 ~~M2~~ S !

a) S = 0; i = 1 ; while (i<10) { S = S*(i+1) ; i++; }

$\underbrace{S=0}_{S0=}$ $\underbrace{i=1}_{i1=}$ $\underbrace{i<10}_{i10<}$ $\underbrace{S = S*(i+1)}_{SS\ i1+*=}$ $\underbrace{i++}_{i++}$

S0 = ; i 1 = ; i10 < 24 !F S S i 1 + * = ; 1 i 1 + ; 7 !